VAAGDEVI COLLEGE OF ENGINEERING

Department of Electrical and Electronics Engineering

Course End Project Report

DC Circuit Solver Using Mesh Analysis

Course	A9205 - Basic Electrical Engineering Laboratory (VCE-R25)
Academic Year	2025-2026
Project Type	Programming-Based Analytical Model
Faculty / Course Instructor	RAVIVARMAN SHANMUGASUNDARAM

S. No	Student Name	Roll Number
1	ANIRUDH UPADHYAY	25881A05Y7
2	MAMIDIPALLY HANSIKA	25881A05Z6
3	AMATHUL LUBNA	25881A05AH

Submitted in partial fulfillment of the Course End Project requirements

Date of Submission: _____

Certificate

Certification

This is to certify that the Course End Project entitled **"DC Circuit Solver Using Mesh Analysis"** is a bonafide work carried out by **ANIRUDH UPADHYAY (25881A05Y7)**, **MAMIDIPALLY HANSIKA (25881A05Z6)**, and **AMATHUL LUBNA (25881A05AH)** in partial fulfillment of the requirements for **A9205 - Basic Electrical Engineering Laboratory (VCE-R25)** during the academic year **2025-2026** under the guidance of **RAVIVARMAN SHANMUGASUNDARAM**.

Course Instructor	Head of Department	Principal
_____	_____	_____
RAVIVARMAN SHANMUGASUNDARAM	Name & Signature	Name & Signature

Acknowledgement

We express our sincere gratitude to our faculty guide **RAVIVARMAN SHANMUGASUNDARAM** for the valuable guidance, motivation, and continuous support provided throughout this project. We also thank the Department of Electrical and Electronics Engineering and the management of Vaagdevi College of Engineering for giving us the opportunity and resources to complete this work successfully. Finally, we thank our classmates and family members for their encouragement during the development of this project.

Table of Contents

Section	Description
1	Abstract
2	Introduction
3	Problem Definition
4	Theory Background
5	Mathematical Formulation
6	Algorithm / Flowchart
7	Program Implementation
8	Sample Input and Output
9	Graphs
10	Result Analysis
11	Conclusion
12	References

1. Abstract

This project presents a console-based Python program for solving DC circuits using the mesh analysis method. Mesh analysis is a systematic technique used to determine unknown loop currents in planar electrical networks by applying Kirchhoff's Voltage Law.

The developed program accepts self-resistance, shared resistance, and source voltage values as input. It automatically forms the mesh equations, solves them, displays the calculated mesh currents, verifies the result using KVL, and generates graphs for visualization.

2. Introduction

DC circuit analysis is one of the fundamental topics in Basic Electrical Engineering. In practical electrical networks, multiple loops may be present, making direct manual calculation lengthy and error-prone. Mesh analysis provides a systematic and efficient method to solve such circuits.

The main objective of this project is to build a programming-based analytical model that solves DC circuits using mesh analysis. Instead of solving the equations manually every time, the program automates the entire process.

3. Problem Definition

The problem is to develop a program that can accept the number of meshes in a DC circuit, accept resistance values of each mesh, accept shared resistance values between adjacent meshes, accept source voltages in each mesh, form the corresponding mesh equations, solve the equations numerically, display the unknown mesh currents with proper units, and verify the correctness of the solution using Kirchhoff's Voltage Law.

Project Outcome

The final solution is a console-based Python tool that builds the mesh matrix, solves the current values numerically, validates the results with KVL, and creates graph outputs for the report.

4. Theory Background

Kirchhoff's Voltage Law states that the algebraic sum of all voltages around any closed loop in an electrical circuit is zero. In mesh analysis, this law is applied separately to each independent loop.

- A mesh current is a hypothetical current assigned to a loop in a planar circuit.
- All mesh currents are assumed clockwise in this project.
- Diagonal terms in the matrix contain self resistance values.
- Off-diagonal terms contain the negative of shared resistance values.

5. Mathematical Formulation

Matrix Equation

For an n-mesh circuit, the equations are written in matrix form as **[R][I] = [V]**, where [R] is the resistance matrix, [I] is the mesh current vector, and [V] is the source voltage vector.

Two-Mesh Equation Set

```
R11*I1 - R12*I2 = V1
-R21*I1 + R22*I2 = V2
```

Assumptions used in this project are: the circuit is planar, all mesh currents are assumed clockwise, the input values are ideal numerical values, resistance is measured in ohms, voltage in volts, and current in amperes.

6. Algorithm / Flowchart

- Start the program and display the title.
- Choose sample mode or manual input mode.
- Read self resistance, shared resistance, and source voltage values.
- Form the resistance matrix.
- Solve the linear equations using Gaussian elimination.
- Display mesh currents and shared-branch current.
- Verify the solution using KVL.
- Generate graphs and end the program.

7. Program Implementation

The project is implemented in Python as a console-based application. Important features include input validation, automatic construction of the resistance matrix, Gaussian elimination for solving simultaneous equations, readable display of mesh equations, KVL verification after solving, and graph generation using matplotlib.

Main Source File

The complete program is stored in **src/dc_mesh_analysis_solver.py**. Important functions include read_int(), read_float(), build_matrix(), solve_linear_system(), print_equations(), verify_kvl(), and generate_graphs().

8. Sample Input and Output

Sample Parameter	Value
Self Resistance of Mesh 1	6 ohms
Self Resistance of Mesh 2	8 ohms

Shared Resistance	2 ohms
Source Voltage of Mesh 1	10 V
Source Voltage of Mesh 2	5 V

Sample Output Obtained from the Program

```
Mesh equations formed from the input data:
Mesh 1: 6.000 * I1 -2.000 * I2 = 10.000 V
Mesh 2: -2.000 * I1 +8.000 * I2 = 5.000 V

Calculated mesh currents:
I1 = 2.0455 A (clockwise)
I2 = 1.1364 A (clockwise)

Current through shared branches:
Between Mesh 1 and Mesh 2: 0.9091 A

KVL verification:
Mesh 1: Left side = 10.0000 V, Right side = 10.0000 V, Error = 0.000000
Mesh 2: Left side = 5.0000 V, Right side = 5.0000 V, Error = -0.000000
```

9. Graphs

The following graphs were generated from the project program and included to satisfy the CEP requirement for graphical output with proper title, axis labels, units, and grid.

9.1 Bar Graph of Mesh Currents

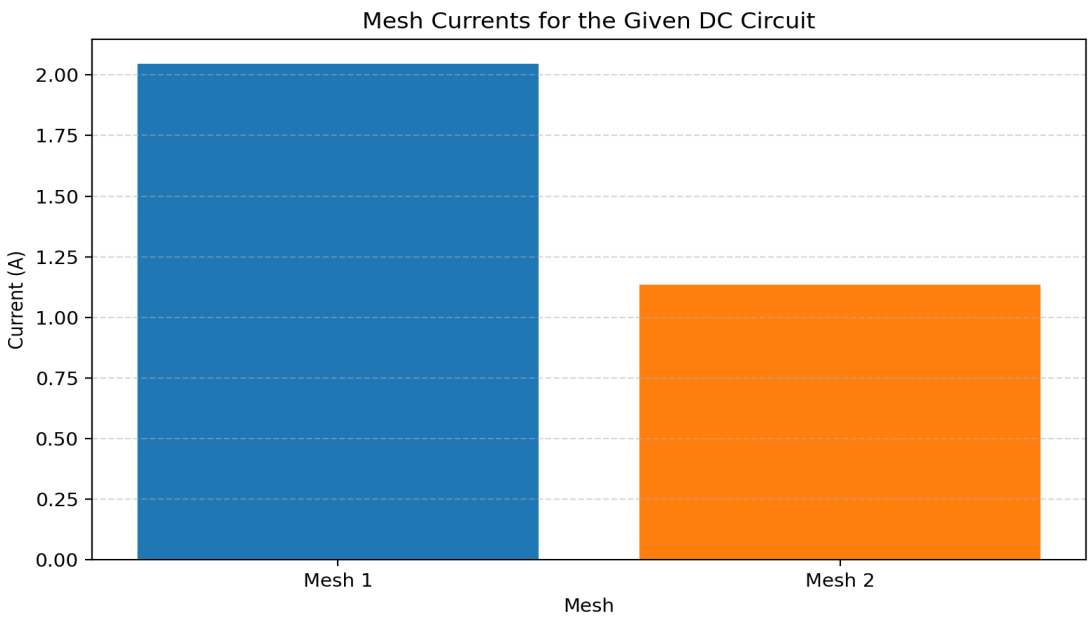

Figure 1: Comparison of mesh current magnitudes for the selected circuit.

9.2 Variation of Mesh 1 Current with Source Voltage

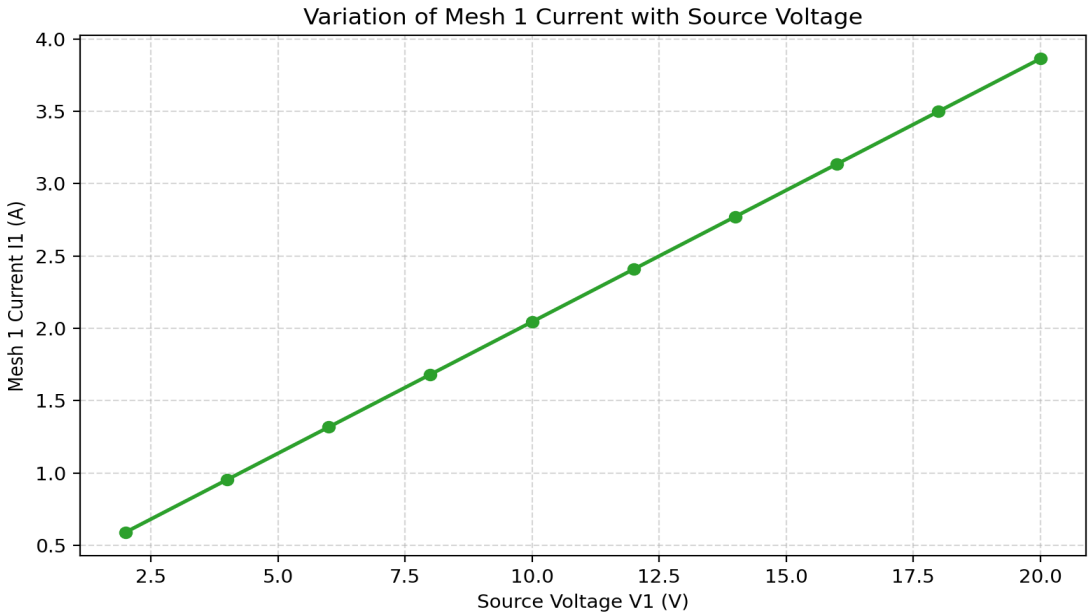

Figure 2: Linear variation of Mesh 1 current with increasing source voltage V1.

10. Result Analysis

The developed program successfully solves DC mesh equations using numerical methods. The computed values are accurate and satisfy the original KVL equations. This confirms the correctness of both the mathematical formulation and the implementation logic.

The project demonstrates that computational methods are very useful when circuits have multiple meshes. Instead of solving simultaneous equations manually, the program handles matrix formation, elimination, validation, and graph generation automatically. This reduces effort and improves reliability.

11. Conclusion

The project **DC Circuit Solver Using Mesh Analysis** successfully fulfills the objectives of the Course End Project for Basic Electrical Engineering Laboratory. The program converts electrical theory into a structured computational model and provides accurate mesh current solutions for DC circuits.

This project helped in understanding Kirchhoff's Voltage Law, matrix representation of circuit equations, and numerical solution methods. It also showed how programming can be effectively used in engineering analysis.

12. References

- Prescribed Basic Electrical Engineering textbook.
- Class notes on Kirchhoff's Laws and mesh analysis.
- Laboratory manual for A9205 - Basic Electrical Engineering Laboratory.

Appendix: How to Run

Run the program using **pip install -r requirements.txt** followed by **python src\dc_mesh_analysis_solver.py**.